

CS407 In Class Activity

Linear Regression and Elastic Net

Your group will investigate how one controlled data variation affects regression results. You will fit ordinary linear regression and a scaled Elastic Net model, evaluate both with the same procedure used by every group, and explain what changed.

Synthetic instructional data All records in this activity were generated for classroom use and do not represent actual students. Associations do not establish that study, sleep, or attendance changes cause higher scores. Treat the results as model evidence within this synthetic sample, not as recommendations about individuals.

Learning goals

- Build and evaluate a multiple linear regression model.
- Explain why feature scaling matters when coefficients are penalized.
- Compare ordinary least squares with Elastic Net regularization.
- Identify how outliers, irrelevant features, and subgroup distributions affect interpretation.

Dataset assignments

Group	Team	Dataset	Variation
1	PEM	Group1_Baseline.csv	Clean baseline
2	Team Terminal	Group2_Sleep_Outlier.csv	One unusual Sleep_Hours observation
3	OMAC	Group3_Music_Genre.csv	Additional categorical feature
4	SKA	Group4_Distance.csv	Additional numeric feature
5	Black Eagle	Group5_Department_Shift.csv	Department-based predictor distributions

Tasks

- 1. Define the modeling problem** Use Exam_Score as the target y . Identify the candidate predictors X and their data types. Exclude Synthetic_Record_ID from every model.
- 2. Inspect data quality** Report the number of records and fields, missing values, duplicate identifiers, plausible ranges, summary statistics, and any unusual observations. Examine relationships with plots or correlations that are appropriate for each data type.
- 3. Prepare the predictors** Encode categorical variables with one reference category, using OneHotEncoder with drop='first' and handle_unknown='ignore'. Put all preprocessing inside each modeling pipeline so it is fitted only on the training portion of each fold.
- 4. Fit ordinary linear regression** Fit an unscaled ordinary least-squares model to obtain the fitted equation and coefficients in the original units. Also fit a standardized ordinary least-squares version for coefficient comparison. Scaling changes coefficient units but not ordinary least-squares predictions when applied correctly.
- 5. Fit scaled Elastic Net** Use StandardScaler followed by ElasticNetCV. Tune alpha and l1_ratio within the training data. Record the selected values and identify coefficients that are strongly reduced or set to zero.
- 6. Evaluate and compare** Use the common evaluation procedure below. Compare mean performance and variability rather than selecting a model from a single split or from training R-squared.
- 7. Interpret the variation** Explain how your group's dataset variation affected coefficients, residuals, stability, or estimated generalization. State one limitation and give one non-causal interpretation of an observed association.

Models to retain

Build and retain all three models because each serves a different purpose. Use the unscaled linear regression model to interpret coefficients in their original units. Use the standardized linear regression model to compare coefficient magnitudes on a common scale. Use the scaled Elastic Net model to examine how regularization shrinks or removes coefficients and whether it improves cross-validated performance.

Model	Preprocessing	Purpose
Unscaled linear regression	One-hot encoding only	Fitted equation and raw-unit coefficients
Standardized linear regression	One-hot encoding then StandardScaler	Standardized OLS coefficient comparison
Scaled Elastic Net	One-hot encoding then StandardScaler	Regularized prediction and coefficient shrinkage

Common evaluation procedure

Every group must use the settings below. Keeping the evaluation procedure fixed allows the class to attribute differences to the datasets rather than to different random splits.

Item	Required setting
Outer evaluation	RepeatedKfold with 5 splits, 5 repeats, shuffle enabled, and random_state=407
Elastic Net tuning	ElasticNetCV with 5 inner folds on each outer training set
Candidate l1_ratio values	0.1, 0.5, and 0.9
Metrics	Mean and standard deviation of MAE, RMSE, and R-squared across the 25 outer folds
Prediction plot	Use one separate shuffled 5-fold KFold with random_state=407 and cross_val_predict
Leakage rule	Fit encoding, scaling, and Elastic Net tuning inside the pipeline, never on the full data before validation

Presentation deliverables

- Target and predictor definitions
- Your dataset's special variation and basic data-quality findings
- Ordinary linear regression and scaled Elastic Net results
- Mean and standard deviation of MAE, RMSE, and R-squared using the common procedure
- A predicted-versus-actual or residual plot based on cross-validated predictions
- Raw versus standardized coefficient interpretation
- An explanation of how the group-specific variation affected the model
- One limitation and one non-causal interpretation statement

Group specific questions

Group	Question to answer
1	Which relationships and coefficient rankings are stable in the baseline data?
2	How does the unusual sleep value affect the Sleep_Hours coefficient, residuals, and validation results?
3	Do the music categories improve validation performance, and how does Elastic Net treat their coefficients?
4	Is distance associated with the target or other predictors, and what happens when it is added to each model?
5	How do predictor distributions differ by department, and does Department add information after the other predictors are included?

Presentation length Plan for approximately ten minutes, followed by up to five minutes of questions.